Proportional Reasoning-1 – Homework Answer Key

CBSE Class 8 Mathematics · Ganita Prakash (Part 1) · Chapter 7

Section A – Ratios

Q1. There are 6 circles and 4 triangles (10 shapes). (a) 6 : 4 = **3 : 2** (b) 4 : 10 = **2 : 5**

Q2. (a) 2 : 3 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2

Q3. (a) 18 : 12 = **3 : 2** (b) Total = 30, so 12 : 30 = **2 : 5**

Q4. ₹2 = 200 paise. 50 : 200 = **1 : 4**

Q5. 2 kg = 2000 g. 400 : 2000 = **1 : 5**

Q6. 2 : 5 = 6 : **15** = **8** : 20 = 10 : **25**

Q7. 2 shaded and 3 unshaded parts (5 in all). (a) **2 : 3** (b) **2 : 5**

Section B – Proportion and Resizing Pictures

Q8. (a) Yes (9 : 12 = 3 : 4) (b) No (4 : 9 cannot be simplified to 2 : 5) (c) Yes (both are 3 : 4)

Q9. (a) 20 = 5 × 4, so x = 3 × 4 = **12**. (b) 12 = 4 × 3, so x = 27 ÷ 3 = **9**.

Q10. 25 : 40 = 5 : 8 and 30 : 48 = 5 : 8. The ratios are equal, so **yes**, they are in proportion.

Q11. A: 6 : 4 = 3 : 2, B: 9 : 6 = 3 : 2, C: 8 : 5. So **A and B** have the same ratio.

Q12. Width 12 → 18 is multiplied by 1.5 (or: height is 2/3 of width). Height = 8 × 1.5 = **12 cm**.

Q13. Width : height = 20 : 15 = 4 : 3. New width 8 = 4 × 2, so height = 3 × 2 = **6 cm**.

Q14. (a) No. 6 : 4 = 3 : 2, but 9 : 8 is not 3 : 2. (b) Yes, it will look stretched (too tall). (c) 9 : h = 3 : 2, so h = **6 cm**.

Section C – Rule of Three

Q15. 1 pen costs 60 ÷ 4 = ₹15. 7 pens cost 7 × 15 = **₹105**.

Q16. 1 kg costs 300 ÷ 5 = ₹60. 8 kg cost 8 × 60 = **₹480**.

Q17. In 1 hour: 120 ÷ 3 = 40 km. In 5 hours: 5 × 40 = **200 km**.

Q18. 1 notebook costs 480 ÷ 12 = ₹40. Number for ₹200 = 200 ÷ 40 = **5 notebooks**.

Q19. 6 people → 2 cups, so 3 people → 1 cup. 15 people → 15 ÷ 3 = **5 cups**.

Q20. 7 × 5 = **35 km**.

Q21. Height : shadow = 2 : 3 for both. 12 = 3 × 4, so height = 2 × 4 = **8 m**.

Q22. In 1 minute: 150 ÷ 5 = 30 bottles. In 12 minutes: 12 × 30 = **360 bottles**.

Section D – Sharing, but Not Equally

Q23. Total parts = 2 + 3 = 5. 1 part = 500 ÷ 5 = ₹100. Asha = 2 × 100 = **₹200**, Ravi = 3 × 100 = **₹300**.

Q24. Total parts = 7. 1 part = 35 ÷ 7 = 5. Tina = 3 × 5 = **15**, Sonu = 4 × 5 = **20**.

Q25. 4 parts = 20 boys, so 1 part = 5. Girls = 5 × 5 = **25**. Total members = 20 + 25 = **45**.

Q26. Total parts = 1 + 2 + 3 = 6. 1 part = 600 ÷ 6 = ₹100. Meena = **₹100**, Kabir = **₹200**, Joseph = **₹300**.

Q27. Total parts = 5. 1 part = 20 ÷ 5 = 4 L. Blue = 2 × 4 = **8 L**, Yellow = 3 × 4 = **12 L**.

Section E – Unit Conversion

Q28. (a) 4 × 12 = **48 inches** (b) 60 ÷ 12 = **5 feet**

Q29. 5 × 84 = **₹420**.

Q30. 40 × 2.5 = **100 cm**.